Learning to Speak as Someone: EM-Style Training for Personality-Conditioned Language Models at Scale

Master Thesis Project Description — Stream A (Persona Only)

Motivation

Large language models are becoming a default tool for simulating human responses — from synthetic survey data to educational dialogue to policy analysis. Yet recent work has exposed a fundamental problem: these models produce an *artificial hivemind*. Across 70+ models, outputs converge on strikingly similar responses that flatten the full spectrum of human personality into a narrow consensus (Jiang et al. 2025). Alignment procedures like RLHF (reinforcement learning from human feedback — the process of training models to follow human preferences) are the primary driver, compressing output diversity far below what even the base models produce (Kirk et al. 2024; Padmakumar and He 2024). When conditioned on demographic descriptions, LLMs approximate population averages but systematically underestimate variance — producing stereotypes, not people (Bisbee et al. 2024).

This thesis asks: can we train a language model that preserves the diversity of real human behavior by conditioning it on individual identity?

Background

A growing body of work attempts to personalize language models to individual users. The dominant paradigm is *embedding-based conditioning*: a user's history is compressed into a learned vector — a continuous numerical representation prepended to the model's input — that steers generation from a frozen model. PERSOMA (Hebert et al. 2024) and the User Embedding Model (Doddapaneni et al. 2024) compress text histories into such soft prompt embeddings. Embedding-to-Prefix (Huber et al. 2025), developed at Spotify, projects pre-computed user embeddings into a single soft token for production-scale personalization. PPlug (Liu et al. 2024) trains a lightweight embedder module that maps historical text into persona vectors. At the adapter level, OPPU (Tan et al. 2024) assigns each user a dedicated LoRA module (a parameter-efficient fine-tuning method that modifies a small fraction of model weights), while Profile-to-PEFT (Tan et al. 2025) uses a hypernetwork to generate per-user adapter weights in a single forward pass. A recent survey covers the full landscape (Zhang et al. 2025).

All these approaches keep the language model backbone frozen and only train the conditioning mechanism. None combines backbone fine-tuning with embedding optimization in an alternating loop. The closest structural analogue comes from image generation, where Textual Inversion (Gal et al. 2022) optimizes per-concept embeddings against a frozen model and DreamBooth (Ruiz et al. 2023) fine-tunes the full model with identity tokens — but this alternation has not been applied to text LLMs. In self-training, ReST^{EM} (Singh et al. 2024) frames iterative fine-tuning as an EM algorithm but operates on discrete outputs rather than continuous persona representations.

A second gap concerns data. Existing personalization work uses recommendation interactions (movie ratings, product reviews) or synthetic personas. Character-LLM (Shao et al. 2023) fine-tunes on individual identities but only for a handful of historical figures. No existing system learns per-person embeddings from thousands of real speakers' natural language. This thesis fills both gaps: it proposes the first EM-style alternating training protocol for persona embeddings and applies it to speech transcripts from thousands of identified speakers at scale.

Core Idea

We propose a *personality-conditioned language model* — a model that learns a dedicated embedding (a learned numerical vector that represents a person's behavioral signature) for each individual speaker in the training data. Given a persona embedding, the model generates text that reflects that specific person's style, vocabulary, beliefs, and reasoning patterns.

The key methodological insight comes from an analogy to the Expectation-Maximization (EM) algorithm, a classical technique for learning when some variables are unobserved. In our case, the persona embeddings are the unobserved variables. Standard fine-tuning trains the model and the embeddings jointly in a single pass. We instead propose an *alternating training protocol* that decouples the two:

- **Phase A (model update):** Fix the persona embeddings and fine-tune the model weights to generate text conditioned on them — standard supervised fine-tuning (SFT).
- **Phase B (embedding update):** Freeze the model and optimize each person's embedding against their data — essentially soft prompt tuning (Lester, Al-Rfou, and Constant 2021; Li and Liang 2021) applied per individual.

This alternation is theoretically grounded. The two time-scale update rule (Heusel et al. 2017) guarantees convergence when components are updated at different rates. Block coordinate descent theory (Xu, Bao, and Xing 2023) provides formal convergence guarantees for alternating minimization in deep networks. Empirically, iterative self-training loops show that 2–3 rounds of alternation capture most of the benefit (Singh et al. 2024). Noise injection during the embedding update phase prevents degenerate convergence (Xie et al. 2020).

The approach sits at the intersection of two established paradigms: prompt tuning (which freezes the model and trains only the input) and conditional SFT (which trains the full model with fixed conditioning). Neither community has combined these into an alternating loop for entity-level personalization at scale. The diffusion model literature does something structurally analogous — Textual Inversion (Gal et al. 2022) optimizes embeddings against a frozen model, DreamBooth (Ruiz et al. 2023) fine-tunes the full model — but the explicit alternation and its EM framing are new for text LLMs.

Beyond novelty, there is a practical reason why alternation may be essential rather than optional: **per-person data is scarce.** A celebrity podcast host may have hundreds of hours of speech, but most guests appear once for 30 minutes. In joint training, a guest's persona embedding receives very few gradient updates before the model has moved on — not enough to learn a meaningful representation. The EM protocol addresses this asymmetry directly. Phase A trains the model on *everyone's* data (millions of examples), so it becomes very good at using persona information in general. Phase B then asks a well-defined question for each individual: "given this already-capable model, what embedding vector best explains this person's 30 minutes of speech?" This is a much easier optimisation problem than jointly learning both from scratch — analogous to how Textual Inversion can learn a concept from just 3–5 images precisely because the generative model is already frozen and capable (Gal et al. 2022). The long tail of speakers with limited data is exactly where the interesting diversity lives, and it is where EM should matter most.

Research Question

Which EM training protocol produces the best persona embeddings for personality-conditioned language models at scale?

The baseline is naive SFT — persona tokens trained jointly with model weights in a single pass. We systematically compare EM protocols across five axes: alternation frequency, phase ordering, asymmetry (the ratio of embedding updates to model updates), number of rounds, and regularisation strategy. The comparison is conducted on a dataset of thousands of real speakers, at a scale where failure modes like persona collapse (all embeddings converging to the same vector) and cold-start degradation (poor performance for speakers with little data) become visible.

Method

This section describes how the training works in practice. The details will be refined during the project — what follows is the starting point.

Training format

Each training example is a segment of text preceded by a special persona token — a placeholder that tells the model "this text was produced by speaker X." For example:

```
[SPEAKER_4217] I think the fundamental issue here is not whether
we regulate, but how we ensure that regulation doesn't stifle the
very innovation we're trying to protect...
```

The persona token is not a fixed string — it is a *learned embedding.* At the start of training, each speaker's embedding is initialised randomly. Over the course of training, the embeddings are optimised so that the model produces text matching each speaker's actual behavior when conditioned on their token.

The alternating training loop

Training proceeds in rounds. Each round has two phases:

Phase A — Train the model. All persona embeddings are frozen. The model weights are updated via standard next-token prediction: given a persona embedding and the preceding tokens, predict the next token. This is ordinary supervised fine-tuning (SFT), except the persona embedding is held constant. The model learns *how to use* persona information for generation, without changing what the personas represent.

Phase B — Train the embeddings. The model weights are frozen. Each speaker's persona embedding is updated by gradient descent on that speaker's data: adjust the embedding so that the frozen model better predicts this speaker's text. This is equivalent to soft prompt tuning (Lester, Al-Rfou, and Constant 2021) applied independently per person. Embeddings are updated with a higher learning rate than the model (following the two time-scale principle (Heusel et al. 2017)) and with noise injection to prevent all embeddings from collapsing to the same vector (Xie et al. 2020).

One round of (A $\rightarrow$ B) takes the same compute as a single pass of standard SFT. We expect 2–3 rounds to suffice (Singh et al. 2024), so total training cost is 2–3x a naive baseline.

What we compare

The core experiment is: does this alternating protocol produce better persona embeddings than training everything jointly? We compare:

- **Baseline (naive SFT):** Persona embeddings and model weights trained jointly in a single pass. This is what you get if you just add speaker tokens to the training data and fine-tune normally.
- **EM alternation:** The two-phase protocol described above. We vary the protocol along five axes — how often to alternate, which phase goes first, how many embedding steps per model step, how many rounds, and what regularisation to apply — to find the best recipe.
- **Additional baselines:** Embedding-only training against a frozen model (pure prompt tuning), and model-only SFT with speaker names as plain text (no learned embeddings).

The comparison is run first at small scale (100 speakers) to narrow the search, then at full scale (thousands of speakers) to confirm.

Data

We build on an existing lab asset: approximately 1 million hours of transcribed podcasts and YouTube videos. From this corpus, we extract a benchmark dataset of thousands of identifiable speakers with varying data volumes — from celebrity hosts with hundreds of hours to one-time guests with minutes. Each training example is a segment of speech paired with a speaker identity tag. The dataset is released as a benchmark for personalized LLM research.

Data pipeline

The pipeline transforms raw transcripts into structured training examples. It is designed to be largely automated, leveraging coding agents to accelerate development.

Step 1 — Source discovery. The existing corpus covers diverse podcasts and YouTube channels, but not all sources contain identifiable speakers. A coding agent scans the corpus metadata, identifies episodes with guest information (show notes, video descriptions, channel metadata), and produces a ranked list of sources by speaker identifiability and data volume.

Step 2 — Speaker extraction and linking. For each source, the pipeline extracts speaker identities from episode metadata and links them across appearances. A podcast guest who appears on three different shows should map to a single speaker ID. This involves fuzzy name matching, cross-referencing against public databases (Wikipedia, LinkedIn, podcast directories), and LLM-assisted disambiguation ("Is 'Dr. Sarah Chen' on episode 47 the same person as 'Sarah Chen, MIT' on episode 112?"). The output is a speaker registry: a table mapping each speaker ID to their canonical name, known appearances, and approximate data volume.

Step 3 — Diarisation and segmentation. Each transcript is segmented into speaker turns using existing diarisation output (many transcription services provide this) or re-processed where needed. Long monologues are split into segments of roughly 500–2000 tokens — long enough to carry stylistic signal, short enough for efficient training.

Step 4 — Mapping to training format. A coding agent writes the transformation from each source's raw format into the unified training schema: `[SPEAKER_ID] <text>`. Different sources have different transcript formats (SRT, VTT, plain text, JSON); the agent writes a source-specific adapter for each, validated against a small manually checked sample. Once an adapter passes validation, it runs over the full source unsupervised.

Step 5 — Quality filtering. Automated checks remove: segments shorter than 50 tokens, segments with high transcription error rates (detected by perplexity under a small language model), segments where speaker attribution is uncertain, and speakers with fewer than 5 segments total. The filtering criteria are defined once; execution is fully automated.

The key insight is that this pipeline is *embarrassingly parallelisable* across sources. Each source is independent: discover format → write adapter → validate → run. Coding agents handle the adapter-writing step, which was historically the bottleneck. The student's role is defining the training schema, setting quality thresholds, and curating the speaker registry — not writing format parsers by hand.

Timeline

Month 1 — Data model and pipeline. Define the training format, build speaker extraction and diarisation pipeline, prototype on a subset of the existing corpus.

Month 2 — Scale data collection. Run the pipeline on the full corpus, debug, curate speaker identities, build the benchmark dataset.

Month 3 — Training infrastructure. Implement the EM training loop, baseline SFT, and evaluation harness. Run first small-scale experiments (100 speakers). Data collection continues in the background.

Month 4 — Full-scale training. Train the model on thousands of speakers using the best protocol candidates from small-scale experiments. 8x H200/B200 GPUs.

Month 5 — Evaluation. Held-out perplexity, speaker identification accuracy, persona collapse detection, cold-start analysis.

Month 6 — Writing. Thesis, optional blog post or demo.

Evaluation

Two primary metrics, chosen for clarity and scalability:

1. **Held-out perplexity per speaker.** For each speaker, measure how well the model predicts held-out text when conditioned on their embedding versus a generic embedding. The gap quantifies how much individual signal the embedding captures.
2. **Persona collapse detection.** Measure the ratio of within-cluster to between-cluster variance in the embedding space. If all embeddings converge (collapse), the model has failed to personalise. This metric tracks whether the EM protocol preserves individual diversity.

Secondary metrics for the thesis include speaker identification accuracy from generated text and cold-start performance (speakers with fewer than 50 utterances).

Expected Outcome

The thesis delivers:

- A large-scale benchmark dataset of personality-conditioned speech data
- A trained personality-conditioned language model
- An empirical comparison of EM training protocols with actionable recommendations

The core principle — that alternating training produces better persona embeddings than joint training — is the hypothesis to be validated. Even if the effect is modest, the dataset and trained model are valuable assets for follow-up work.

Path to Publication

With 3–4 additional months of work after the thesis, the project targets a NeurIPS submission. The publication would expand the evaluation (scaling curves, compute efficiency analysis, interaction effects across protocol dimensions) and position the contribution as a validated recipe for persona embedding training. The trained model and persona embeddings also serve as the foundation for a Nature-track paper investigating whether personality-conditioned models can recover the human behavioral diversity that alignment erases (Jiang et al. 2025; Shumailov et al. 2024; Wu, Black, and Chandrasekaran 2024).

References

Bisbee, James, Joshua D. Clinton, Cassy Dorff, Brenton Kenkel, and Jennifer M. Larson. 2024. "Synthetic Replacements for Human Survey Data? The Perils of Large Language Models." *Political Analysis* 32: 401–16. https://doi.org/10.1017/pan.2024.5.

Doddapaneni, Sumanth, Krishna Sayana, Ambarish Jash, Sukhdeep Sodhi, and Dima Kuzmin. 2024. "User Embedding Model for Personalized Language Prompting," January.

Gal, Rinon, Yuval Alaluf, Yuval Atzmon, Or Patashnik, Amit H. Bermano, Gal Chechik, and Daniel Cohen-Or. 2022. "An Image Is Worth One Word: Personalizing Text-to-Image Generation Using Textual Inversion," August.

Hebert, Liam, Krishna Sayana, Ambarish Jash, Alexandros Karatzoglou, Sukhdeep Sodhi, Sumanth Doddapaneni, Yanli Cai, and Dima Kuzmin. 2024. "PERSOMA: PERsonalized SOft ProMpt Adapter Architecture for Personalized Language Prompting." In *Proceedings of Workshop on Generative AI in Recommendation and Personalization at KDD '24 (GenAIRecP @ KDD'24)*.

Heusel, Martin, Hubert Ramsauer, Thomas Unterthiner, Bernhard Nessler, and Sepp Hochreiter. 2017. "GANs Trained by a Two Time-Scale Update Rule Converge to a Local Nash Equilibrium." In *Advances in Neural Information Processing Systems*. Vol. 30. https://arxiv.org/abs/1706.08500.

Huber, Bernd, Ghazal Fazelnia, Andreas Damianou, Sebastian Peleato, Max Lefarov, Praveen Ravichandran, Marco De Nadai, Mounia Lalmas-Roellke, and Paul N. Bennett. 2025. "Embedding-to-Prefix: Parameter-Efficient Personalization for Pre-Trained Large Language Models." https://arxiv.org/abs/2505.17051v1.

Jiang, Liwei, Yuanjun Chai, Margaret Li, Mickel Liu, Raymond Fok, Nouha Dziri, Yulia Tsvetkov, Maarten Sap, Alon Albalak, and Yejin Choi. 2025. "Artificial Hivemind: The Open-Ended Homogeneity of Language Models (and Beyond)." In *39th Conference on Neural Information Processing Systems (NeurIPS 2025)*.

Kirk, Robert, Ishita Mediratta, Christoforos Nalmpantis, Jelena Luketina, Eric Hambro, Edward Grefenstette, and Roberta Raileanu. 2024. "Understanding the Effects of RLHF on LLM Generalisation and Diversity." In *International Conference on Learning Representations (ICLR)*.

Lester, Brian, Rami Al-Rfou, and Noah Constant. 2021. "The Power of Scale for Parameter-Efficient Prompt Tuning." In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, 3045–59. https://doi.org/10.18653/v1/2021.emnlp-main.243.

Li, Xiang Lisa, and Percy Liang. 2021. "Prefix-Tuning: Optimizing Continuous Prompts for Generation," January.

Liu, Jiongnan, Yutao Zhu, Shuting Wang, Xiaochi Wei, Erxue Min, Yu Lu, Shuaiqiang Wang, Dawei Yin, and Zhicheng Dou. 2024. "LLMs + Persona-Plug = Personalized LLMs," September.

Padmakumar, Vishakh, and He He. 2024. "Does Writing with Language Models Reduce Content Diversity?" In *International Conference on Learning Representations (ICLR)*. Vol. ICLR 2024.

Ruiz, Nataniel, Yuanzhen Li, Varun Jampani, Yael Pritch, Michael Rubinstein, and Kfir Aberman. 2023. "DreamBooth: Fine Tuning Text-to-Image Diffusion Models for Subject-Driven Generation." In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 22500–22510. https://arxiv.org/abs/2208.12242.

Shao, Yunfan, Linyang Li, Junqi Dai, and Xipeng Qiu. 2023. "Character-LLM: A Trainable Agent for Role-Playing." *arXiv Preprint arXiv:2310.10158v2 [Cs.CL]*, December. https:

//arxiv.org/abs/2310.10158v2.

Shumailov, Ilia, Zakhar Shumaylov, Yiren Zhao, Yarin Gal, Nicolas Papernot, and Ross Anderson. 2024. "The Curse of Recursion: Training on Generated Data Makes Models Forget," April.

Singh, Avi, John D Co-Reyes, Rishabh Agarwal, Ankesh Anand, Piyush Patil, Xavier Garcia, Peter J. Liu, et al. 2024. "Beyond Human Data: Scaling Self-Training for Problem-Solving with Language Models." *Transactions on Machine Learning Research*, April.

Tan, Zhaoxuan, Qingkai Zeng, Yijun Tian, Zheyuan Liu, Bing Yin, and Meng Jiang. 2024. "Democratizing Large Language Models via Personalized Parameter-Efficient Fine-Tuning." *arXiv Preprint arXiv:2402.04401*. https://arxiv.org/abs/2402.04401.

Tan, Zhaoxuan, Zixuan Zhang, Haoyang Wen, Zheng Li, Rongzhi Zhang, Pei Chen, Fengran Mo, et al. 2025. "Instant Personalized Large Language Model Adaptation via Hypernetwork." *arXiv Preprint arXiv:2510.16282v1*, October. https://arxiv.org/abs/2510.16282v1.

Wu, Fan, Emily Black, and Varun Chandrasekaran. 2024. "Generative Monoculture in Large Language Models." https://arxiv.org/abs/2407.02209.

Xie, Qizhe, Minh-Thang Luong, Eduard Hovy, and Quoc V. Le. 2020. "Self-Training with Noisy Student Improves ImageNet Classification." In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 10687–98. https://doi.org/10.1109/CVPR42600.2020.01070.

Xu, Jintao, Chenglong Bao, and Wenxun Xing. 2023. "Convergence Rates of Training Deep Neural Networks via Alternating Minimization Methods," April. https://arxiv.org/abs/2208.14318.

Zhang, Zhehao, Ryan A. Rossi, Branislav Kveton, Yijia Shao, Diyi Yang, Hamed Zamani, Franck Dernoncourt, et al. 2025. "Personalization of Large Language Models: A Survey." *Transactions on Machine Learning Research (TMLR)*. https://openreview.net/forum?id=tf6A9EYMo6.